

Raymond Chen

Canadian Citizen • Richmond, BC • Raymondchen085@gmail.com • [LinkedIn](#) • [Portfolio](#)

EXPERIENCE

Teaching Assistant, UBC Mechanical Engineering – Vancouver, BC **Sep – Dec 2025, 2026**

- Led **20+ engineering students** in tutorial sessions on stress analysis, fatigue, and component selection for MECH 325: Machine Design, demonstrating design practices for bearings, belts, shafts, and gears through hands-on problem solving.
- Guided students through design methodology, reinforcing appropriate safety factor and material selection.

Mechanical Manufacturing Engineering Intern, Kardium – Burnaby, BC **Sep 2025 – May 2026**

- Led the implementation of process improvements for manufacturing of biomedical devices in cleanroom environment.
- Designed laser welding fixture to **reduce cycle time** for mandrel manufacturing and **doubling equipment spares**.
- Redesigned jigs in **SolidWorks** to reduce consumable consumption and downtime, **cutting duplication costs by 50%**.
- Created CNC laser-cut parts, engineering drawings and supported time studies, root cause analysis, and process reviews.
- Developed Just-In-Time fabrication process for training proxies, reducing cycle time and cost per operator training.
- **Reduced chemical waste by 90%** leading to a **\$3000/month in cost savings** from designing and implementing new seals.

Mechanical Engineering Intern, LB Foster – Burnaby, BC **Jan 2025 – May 2025**

- Designed mechanical hardware for train-mounted friction management systems for **9+ transit and freight projects**.
- Created SolidWorks mounting tools/jigs and in-house 3D printing processes to **reduce production costs by 60%**.
- Performed **FEA** design validation for fatigue loading and modal analysis per international standards for Rail Vehicle Structures, successfully presenting FEA reports for engineering and client review and approval.
- **Improved component design life by 33%** through dynamic fatigue analysis of random vibrations and shock in CAE Fatigue.
- Drafted manufacturing drawings per GD&T/ANSI/AWS A2.4-98 standards and I&M manuals for client installations.

Systems Engineer Student, AtkinsRéalis – Richmond, BC **May 2024 – Aug 2024**

- Supported operation and maintenance of vehicle assets of the Canada Line, an automated rail rapid transit system.
- Developed Statements of Work (SOWs) for RFPs, specifying upgrade requirements in coordination with external vendors.
- Presented inspection reports to support maintenance planning and budgeting, **identifying \$10K of misallocated funds**.
- **Reduced replacement lead times by 40%** for safety-critical brake hardware by coordinating with OEMs.

Engineering Support Services Coordinator, Ledcor – Vancouver, BC **May 2023 – Dec 2023**

- Directed engineering support for Fibre Home and Business construction, adhering to safety standards and quality control.
- Prepared estimates and BOMs to support budgeting, scheduling, materials planning, and quotations for **\$10M in projects**.
- Designed and reviewed fibre distribution drawings for compliance and constructability in Bluebeam Revu and AutoCAD, facilitating successful client and city approvals across multiple permitting jurisdictions for **100+ service areas**.

EDUCATION

University of British Columbia (UBC) **Expected Graduation: May 2027**

Bachelor's of Applied Science - Mechanical Engineering

CGPA: 87%

Coursework: NC Programming, Machine Design, Manufacturing Processes, Fluid Dynamics, Machine Learning

PROJECTS

Manufacturing Process Design for a MacBook Chassis **Nov 2025 – Dec 2025**

- Designed the vacuum machining fixture and 5-axis milling process for a MacBook, optimizing quality, process time and cost.
- Optimized feeds, speeds, tool selection and toolpaths for NC programming, providing DFM feedback.
- Simulated machining sequences to prevent collisions and refine tool paths and minimize machining time.

Aerodynamics & Composites Team Co-Lead, UBC ThunderBikes **Sep 2023 – Aug 2024**

- Directed fabrication of carbon fiber bodywork parts using wet-layup and vacuum infusion techniques.
- Optimized composite layup techniques, reducing porosity and improving surface finish by iterating fiber and resin combos.
- Reverse-engineered motorcycle bodywork components with **photogrammetry** to design molds for composite layups.

SKILLS

Mechanical Design: SolidWorks, DFAM, FEA, CAM (Siemens NX), Lean Manufacturing, ASME Y14.5 (GD&T)

Manufacturing: 3D Printing (FDM/Resin), Machining (Lathe/Mill), Welding (Laser/MIG/TIG), Water Jet & Laser cutting

Software: Python, C#, MATLAB, Arduino, MSP430, Oscilloscopes